

No-Egress Verification PDF Report

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INTERNAL

EXECUTIVE SUMMARY

Aegis is proposing a sovereign on-premise AI workbench utilizing open-weight multimodal LLMs for confidential industrial work. The system prioritizes data localization, agentic workflows (Plan → Execute → Verify → Re-plan), multimodal document understanding (PDF, Images, Drawings), and secure, sandboxed tool execution. It leverages a dynamic model loading system and focuses on verifiable deliverables with a key emphasis on data sovereignty and control within an organization's network.

KEY FINDINGS

The Sovereign AI Workbench's core functionality revolves around 100% local AI processing, intelligent model routing, and agentic automation. It supports multimodal input (text, PDFs, scans, images, drawings) and offers a modular architecture with a model registry for dynamic loading. A key feature is the dynamic model loading system (1. Task Arrives, 2. Check Required Model, 3. Load If Needed, 4. Execute, 5. Unload If Idle) intended to optimize resource utilization. The system emphasizes verified real-world deliverables and utilizes a private network/on-premise infrastructure to maintain data security and control. It targets applications like inspection report analysis and document preparation, utilizing quantized 7B/8B class LLMs.

DETAILED ANALYSIS

The proposed solution employs a layered architecture: a Secure Web UI for user interaction, an Agent Controller managing task planning, and a Document Workflow for processing PDFs, scans, and text. Coding workflows utilize code generation and execution within a sandboxed environment. Multimodal workflows process images and drawings alongside text and documents. The system incorporates a Model Registry and Manager for selecting and managing AI models. Data resides entirely within the organization's network, eliminating cloud API calls and external data leakage. The 'agentic' approach focuses on automatically executing tasks defined by the 'agent' based on user input and internal knowledge (RAG). The system uses a combination of OCR, Vision models, and text-based LLMs to interpret and process multimodal inputs.

RISKS AND OPERATIONAL ISSUES

Potential risks include GPU limits and contention, requiring careful resource management. The OCR + VLM + Document pipeline presents a complex integration challenge. Hallucination potential of the LLMs and data leakage vulnerabilities necessitate rigorous validation and security protocols. The system relies on a local network/air-gapped deployment which may impact accessibility and collaboration. Maintenance and updates will need to be performed on-site. The system's effectiveness is dependent on the quality and volume of the organization's internal knowledge base (RAG).

RECOMMENDATIONS

To mitigate risks, Aegis should prioritize robust testing and validation of the LLMs’ output, implement detailed audit logs for all agent actions and data access, and establish clear data governance policies. Further development should focus on optimizing the dynamic model loading system to minimize latency and maximize resource utilization. Consideration should be given to developing methods to identify and mitigate LLM hallucination. Establishing a formal process for updating and maintaining the models and infrastructure is crucial for long-term operational viability.

AUTHORITATIVE SOURCES & CITATIONS

Document Name	Referenced Pages	Relevance Status
SIH2026ppt.pdf	Page(s) 1, 2, 3, 4, 5, 6, 7	High

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